

Draft - expert review pending

This unit is complete and passes the platform's automated checks, but no reviewer has read it yet. Treat its sources and claims with care.

Topic 2.3 - System software and application software

A real classroom situation

A pupil in Matiari asks Mr. Waqar a question that stops him: "Sir, when I press the power button, who tells the computer to show the screen? The games and the browser are not open yet, so who is in charge?" The honest answer is that something is in charge from the first instant, before any app exists on the screen: the system software, and above all the operating system. Everything the pupil loves to use - the browser, the game, the photo editor - is a guest in a house the operating system built and runs.

System software and application software

Read each row across: what each type does, when it runs, and what it needs.

Property	System software	Application software
Purpose	Runs and manages the machine itself	Does tasks for the user
Examples	Windows, Linux, Android Phone settings	Word processor, spreadsheet Browser, photo editor
Runs when	From power on to power off	Only when the user opens it
Works with	Hardware directly	Through the operating system
Remove it	Machine stops working	One task disappears

The solid and dashed headers reinforce the written category labels, not a colour-only distinction.

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Explanation

System software and operating systems

System software is the software that runs the machine itself. Its chief member is the operating system (OS). Bourgeois and colleagues (2019) define the operating system's role in three parts: managing the hardware resources of the computer, providing the user-interface components, and providing a platform for software developers to write applications. In plainer words, the OS divides the machine's attention among everything running, draws the screen you interact with, and offers app makers a stable floor to build on.

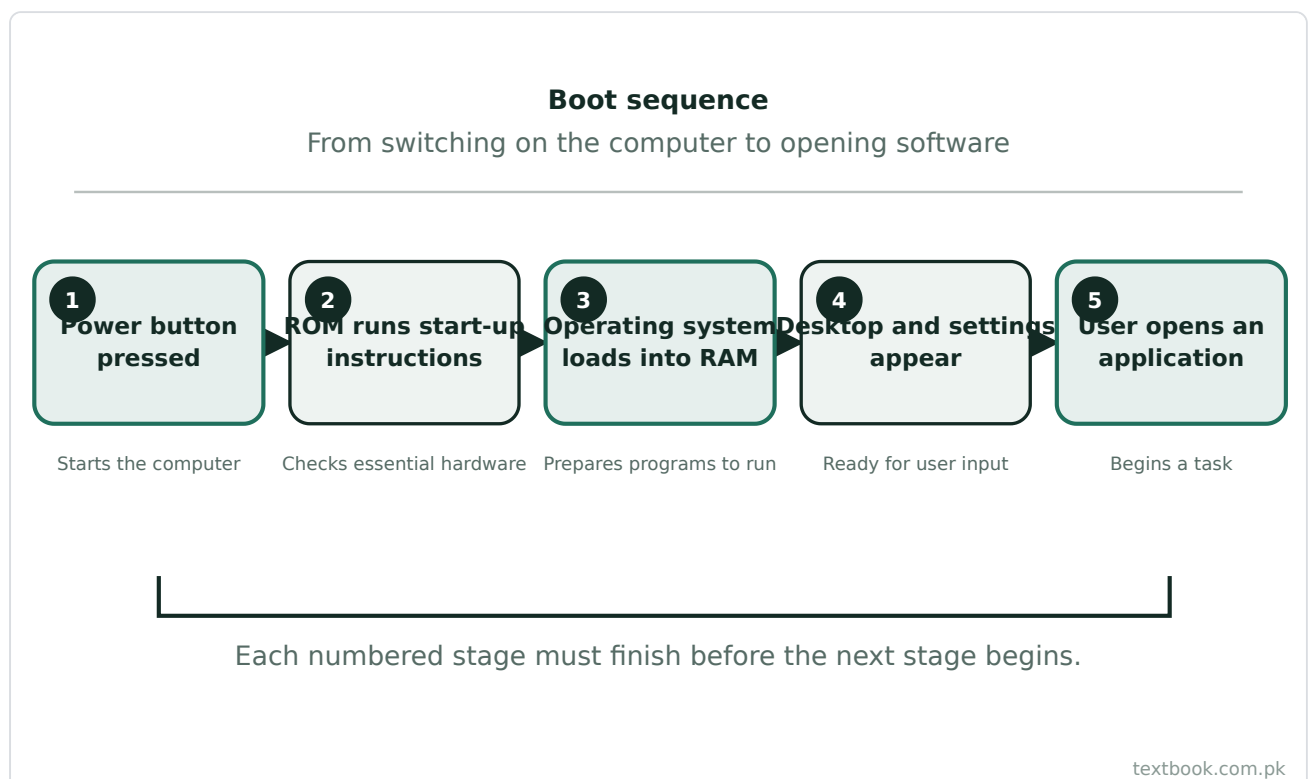
The names you meet daily are operating systems: Microsoft Windows on most office and lab computers; Apple's macOS on Macintosh machines; Linux, which is open source, meaning individual developers are allowed to modify its code, and which quietly runs a large share of the world's servers; Android and iOS on phones (Bourgeois et al., 2019). Every computing device needs one, from the desktop to the phone to the smart TV.

Beside the OS, system software includes utility software: small programs that fix, check or tune the machine - the antivirus scanner, the disk cleanup tool, the backup program. Utilities do not do your work; they keep the machine able to do your work.

Application software and productivity tools

Application software is the software you run to get something done. Bourgeois and colleagues (2019) put it simply: application software "performs specific tasks such as word processing, accounting, database management, video games, or browsing the web". The word processor in which this paragraph would be typed, the spreadsheet holding the class results, the browser, the photo editor, the learning app on a pupil's phone: all applications.

The guide's phrase "productivity tools" names the application family a teacher leans on most: word processing, spreadsheets and presentation software, usually bought together as a suite. Bourgeois and colleagues (2019) note that the spreadsheet was the personal computer's first "killer application" - the app so useful that people bought the machine just to run it - and in a school office that is still visibly true: the attendance register, the marks sheet and the fee ledger all live in spreadsheets.



Now the pupil's question can be answered fully. Press the power button and the machine runs its start-up instructions from ROM; the operating system then loads from the disk into RAM; the desktop appears; only then can any application be opened, and every application you open afterwards asks the operating system for its share of memory, screen and processor time. The boot sequence is the clearest proof of who is in charge: the house opens before any guest arrives.

A caution against a common slip: "software" does not mean only the programs you buy in a box or download. The operating system is software; the utility that backs up the exam disk is software; the school's result-management system is software. The moment you sort by the two questions - does it run the machine, or does it do a task for the user? - every program you meet falls on one side. And do not assume that daily use of software builds real understanding of it: when Wallace and Clariana (2005) tested incoming university students on exactly these concepts, the average score was about 60 percent, well below passing, and only about a third could have exempted the introductory course. Confidence with apps is not knowledge of the machine; that gap is why this unit exists.

Activity: Software census

Work in groups of three. About 20 minutes.

1. List every piece of software you can find evidence of on the classroom computer or your phone (icons, the installed-apps list, the settings screen). Aim for at least twelve.
2. Sort each into system software (operating system, utilities) or application software. For anything you argue about, write down the argument on both sides.
3. Find one item you would call a productivity tool and one you would not, and be ready to defend the difference.
4. As a group, write the two-question test you used (runs the machine vs does a task) and check it against every item once more.

Check your understanding

1. Name the three jobs of an operating system according to this topic's account. *(Remember)*
2. Explain the difference between system software and application software in two sentences, with one example of each. *(Understand)*
3. A colleague says the antivirus program is an application because you open it like any app. Using the system/application distinction, evaluate the claim. *(Apply)*
4. Put these in the order they happen at start-up: desktop appears, ROM instructions run, OS loads into RAM, power button pressed. *(Remember)*

Summary

System software runs the machine: the operating system manages hardware, draws the interface and hosts every app, while utilities keep the machine healthy. Application software does tasks for the user, and the productivity family - word processing, spreadsheets, presentations - carries a school's daily work. From the power button to the desktop, the boot sequence shows the order of authority: ROM starts, the OS loads, and only then do applications run as its guests.

Self-assessment checklist

Sign in to sync your ticks across devices.

- ☐ I can define system software and application software and give three examples of each.
- ☐ I can name the operating system's three jobs.
- ☐ I can narrate the boot sequence from power button to opening an app.

Try this at your practicum school

Ask the lab in-charge which operating system the school's computers run and which three applications are used most. Note the answers, and find out what the most-used application is used for. Bring the list back to class.

Summative task

The head teacher asks you to explain, for a staff meeting, why the school's computers need both the operating system and the applications, and why deleting one is not like deleting the other. Prepare five to eight presentation slides (or one page of notes) that use the boot sequence and the system/application distinction, with at least one everyday school example.

Marking guide (10 marks):

- Both categories defined accurately: 2 marks.
- The boot sequence used as evidence, in correct order: 3 marks.
- The deletion comparison is argued from what each category does, not just asserted (Analyse): 3 marks.
- Fits the audience (staff meeting, plain language): 2 marks.

Further reading

- Bourgeois, D. T., Smith, J. L., Wang, S., & Mortati, J. (2019). *Information Systems for Business and Beyond* (2nd ed.), chapter 3. Open Textbook Library. <https://open.umn.edu/opentextbooks/textbooks/information-systems-for-business-and-beyond>
- Wallace, P., & Clariana, R. B. (2005). Perception versus reality: Determining business students' computer literacy skills and need for instruction in information concepts and technology. *Journal of Information Technology Education: Research*, 4, 141-151. <https://doi.org/10.28945/269>
- Stair, R., & Reynolds, G. *Principles of Information Systems*. Cengage Learning. (Course guide reading; print monograph, cited at bibliographic level.)

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